1 Quantum Error Correction

The main challenge of quantum computation is the fragility of the physical systems used to implement qubits. In classical computation, information is encoded in macroscopic physical variables, and the logical states are separated by energy barriers that make them robust against environmental fluctuations. Compared to classical bits, qubits are heavily affected by environmental noise, which makes them degrade and lose their information due to decoherence. It is thus necessary to explore the most efficient ways to protect quantum information while maintaining the practical feasibility of operating with those protected qubits. [8]

1.1 Overview and Simple Codes

Fathoming the differences between classical and quantum error-corecting codes makes it necessary to pay attention to three difficulties that arise when we deal with quantum systems: The no-cloning theorem,, which prevents us from duplicating the state of a qubit; the continuous nature of errors; and the fact that observation in quantum mechanics collapses the wave function in one of the eigenstates of the measured observable (this is a non-unitary process, and thus recovery is impossible)

In order to overcome these difficulties, quantum error-correcting codes (QECC) utilize entanglement to encode a logical qubit across a multi-qubit system. This way, we can protect quantum information through redundancy, without violating the no-cloning theorem. QEC protocols measure multi-qubit parity operators (error syndromes), avoiding the collapse of the wave function. The continuity problem will be confronted by discretizing the errors into a finite set of Pauli operations.

We will build some basic intuition about quantum error correction models based on redundancy by looking at one of its earliest and simplest examples: the Shor Nine-Qubit Code.

Consider that we have physical qubits that are subject to random bit-flip errors, that is, with a certain probability p, the state $|\psi\rangle$ is transformed into $X|\psi\rangle$. As we cannot simply duplicate our original qubit, we can entangle it with two ancilla qubits initialized in the $|0\rangle$ state by applying two CNOT gates, where the original qubit is the control and the ancilla qubits are the targets. If our original state is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the resulting state will be $|\psi_L\rangle = \alpha|000\rangle + \beta|111\rangle$. The encoding establishes a logical basis with states:

$$|0_L\rangle = |000\rangle$$
$$|1_L\rangle = |111\rangle$$

If the bit-flip error occurs in any qubit, we can diagnose this error without destroying the superposition by measuring the parity observables Z_1Z_2 and Z_2Z_3. Those observables compare the parities of their corresponding qubits, yielding a ±1 eigenvalue, positive if both compared qubits share the same state, negative if they are different. It is crucial to note that the measurement of the parity operators commutes with the logical operators, ensuring that the superposition remains intact. Once we have detected the error, we can apply the appropiate recovery operation.

This same procedure can be adapted to protect against phase-flip errors, modeled by the Pauli Z operator. This is performed by applying a Hadamard transformation to each qubit after applying two sequential CNOT gates. It is an identical procedure, and it only differs in the basis we are working with.

If we concatenate the bit-flip and phase-flip codes, we can protect against an arbitrary error on a single qubit. This is therefore a nine-qubit code, with the base states defined as follows:

$$
\begin{aligned}
|0_L\rangle &= \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle) \\
|1_L\rangle &= \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)
\end{aligned}
$$

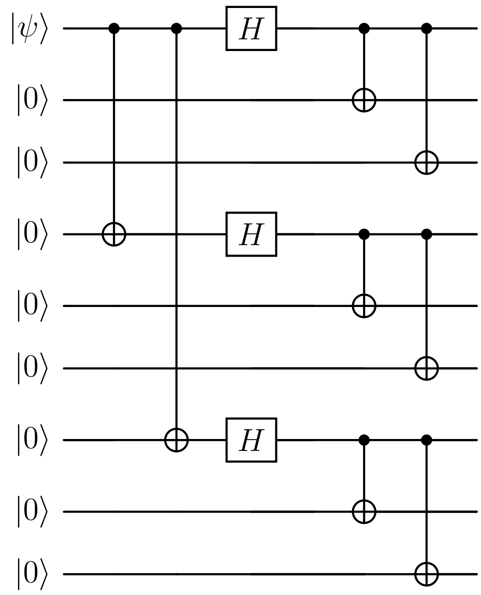

Figure 1: Shor Code circuit

The syndrome measurement is performed in an analogous way. We first compare the first two qubits by applying the operator Z_1Z_2, then we continue comparing each pair of neighboring qubits by applying the corresponding operators. We can detect the bit-flip errors on any of the qubits. In order to detect phase-flip errors, we compare the signs of each of the three-qubit blocks.

The Shor code allows us to detect bit-flip and phase-flip errors. In order to illustrate that this condition is enough to detect an arbitrary error, we will describe the noise by a trace-preserving operation $\mathcal{E}$. Given ρ the density matrix of the system, the noise acting on the state can be expressed as:

$$\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger \tag{1}$$

This is called Kraus operator-sum representation, and will be discussed in more detail. The operators E_k can be expanded as a linear combination of the Pauli matrices $E_k = e_{i0}I + e_{i1}X_1 + e_{i2}Z_1 + e_{i3}X_1Z_1$. The resulting quantum state $E_k |\psi\rangle$ can be expressed as a superposition of $|\psi\rangle$, $X_1 |\psi\rangle$, $Z_1 |\psi\rangle$, and $X_1Z_1 |\psi\rangle$. When we measure the error syndrome, the system collapses to any of the four states, and after applying the necessary correction operation, we obtain the final state $|\Psi\rangle$.

1.2 General Theory of Quantum Error Correction

Once we have built some intuition about QEC, we can construct a more general theory.

Firstly, we need to give a mathematical description of our noise $\mathcal{E}$ [4]. Physically, noise represents an interaction between the system and the environment. Since we do not have access to the environment, the action of noise on the system appears to be a non-unitary operation, and we cannot describe our qubit as if it was in a pure state. This forces us to use the density matrix

$$\rho = \sum_i p_i |\psi_i\rangle \langle\psi_i| \tag{2}$$

that represents the mixture of different pure states $|\psi_i\rangle$ with probabilities p_i. This must be a positive semi-definite Hermitian matrix, with unit trace and non-negative eigenvalues. Non-unitary noise operations must preserve these properties. Hence, we represent noise acting on a system as a transformation of the state described by a density matrix that fulfills this condition: a Completely Positive Trace Preserving (CPTP) map. In the Kraus representation, the action of a CPTP map $\mathcal{E}$ on a density matrix ρ is given by the formula 1, where the Kraus operators E_k satisfy the condition $\sum_k E_k^\dagger E_k = I$ (this condition derives necessarily from the requirement of trace preservation). If we consider any state $|\psi\rangle$ in the Hilbert space and define the states $|\psi_k\rangle = E_k^\dagger |\psi\rangle$, we can easily check that:

$$\sum_k \langle\psi| E_k \rho E_k^\dagger |\psi\rangle = \sum_k \langle\psi_k| \rho |\psi_k\rangle \geq 0$$

which verifies that the map is positive. In fact, Choi's theorem on completely positive maps guarantees that any quantum operation admitting such a Kraus representation is inherently *completely positive*, meaning it yields valid physical states even when applied to a subsystem of a larger entangled system. We will call the operation elements $\{E_i\}$ errors, and they represent the transformation of the quantum state under Hamiltonian parameter deviations from their ideal values.

Now we want to generalize the basic ideas present in the Shor code to give a simple picture of how we can detect and correct errors in a quantum system. We firstly define a quantum error-correcting code as a subspace C of the total Hilbert space, and the elements of C will be called codewords. We encode our quantum states by projecting them into C by a unitary operation P that we will call the projector. The encoded system is then vulnerable to environmental noise, which we model as a CPTP map $\mathcal{E}$. We perform a syndrome measurement to determine the error that has affected the system, and a recovery operation $\mathcal{R}$ returns the system to its original encoded state.

When we project our state into the code subspace, we are essentially restricting the degrees of freedom of the system. This means that if our original Hilbert space for n physical qubits is 2^n-dimensional, the code subspace will be 2^k-dimensional, where $k < n$. We will call C a $[n, k]$ code, as we are encoding k logical qubits into n physical qubits. Once the noise acts on the encoded state $|\psi_L\rangle$, the system will be pushed into a different subspace. Specifically, each error E_α projects the state into an error subspace $C_\alpha = E_\alpha C$. Each error subspace has its corresponding projector $P_\alpha = E_\alpha P E_\alpha^\dagger$.

If, given a code, each error corresponds to different undeformed and orthogonal subspaces, then we can reliably distinguish which error has occurred. This will be achieved by performing the set of projective measurements associated with each error subspace. Given our noisy state $E_\alpha |\psi_L\rangle$, the execution of a projector P_β will yield us a syndrome lecture β with probability $p_\beta = \langle\psi_L| E_\alpha^\dagger P_\beta E_\alpha |\psi_L\rangle$. The orthogonality of error subspaces ensures that the measurement associated to an error subspace C_β will only yield a non-zero probability when the corresponding error has occurred.

The undeformed condition means that, if you have two perfectly distinguishable (orthogonal) states $|\psi_L\rangle$ and $|\phi_L\rangle$ in the code subspace, then the error E_α must not make them non orthogonal. This is equivalent to saying that the inner product of the two states must be preserved under the action of the error, that is, $\langle\psi_L| E_\alpha^\dagger E_\alpha |\phi_L\rangle = \langle\psi_L| |\phi_L\rangle = 0$. This ensures that the recovery operation can correctly restore the original encoded state.

We can define the condition for an error correction code to be able to correct the effect of noise as follows: Given a code C, a noisy quantum channel characterized by $\mathcal{E}$, and any state $|\psi_L\rangle$, with density matrix ρ_L, the error correction is successful if

$$(\mathcal{R} \circ \mathcal{E})(\rho) \propto \rho \tag{3}$$

In this condition, and in the following equation, we have not imposed the trace-preserving condition for $\mathcal{E}$, with the idea of allowing for more general noise models. That is the reason we use the proportionality symbol $\propto$. The prior discussion can be summarized in a very compact algebraic expression, the Knill-Laflamme condition for quantum error correction:

Theorem 1 (Knill-Laflamme)**.** *Let C be a quantum error-correcting code, P the projector onto C, and $\mathcal{E}$ a CPTP map with operation elements $\{E_i\}$. Then, there exists a recovery operation $\mathcal{R}$ such that $(\mathcal{R} \circ \mathcal{E})(\rho) = \rho$ for all ρ supported on C if and only if the following condition holds:*

$$PE_i^\dagger E_j P = \alpha_{ij} P \tag{4}$$

for some Hermitian matrix α_{ij}.

Proof. ($\Leftarrow$) We notice that the Hermitian matrix α_{ij} can be diagonalized by a unitary transformation u, such that $u\alpha u^\dagger = d$, where d is a diagonal matrix. We can define new operation elements $F_i = \sum_j u_{ij} E_j$. Those can be proven to be operation elements of $\mathcal{E}$.

The new operation elements also satisfy the Knill-Laflamme condition, since:

$$
\begin{aligned}
PF_i^\dagger F_j P &= \sum_{k,l} u_{ki}^\dagger u_{jl} P E_k^\dagger E_l P \\
&= \sum_{k,l} u_{ki}^\dagger u_{jl} \alpha_{kl} P \\
&= d_{kl} P
\end{aligned}
$$

which simplifies the Knill-Laflamme condition, since d_{kl} is a diagonal matrix. This diagonalization makes it explicit that the error subspaces are orthogonal. We decompose the operation elements as $F_i P = \sqrt{d_{ii}} U_i P$, where U_i is a unitary operator and $\sqrt{d_{ii}}$ is a positive real number.

F_i maps the code subspace C to the error subspace defined by the projector $P_i = U_i P U_i^\dagger = F_i P U_i^\dagger / \sqrt{d_{ii}}$. As we have already discussed, the syndrome measurement is the projective measurement defined by the projectors P_i. The recovery operation $\mathcal{R}$ is defined as $\mathcal{R}(\rho) = \sum_k U_i^\dagger P_i \rho P_i U_i$. For any state ρ supported on C, we can show that:

$$
\begin{aligned}
U_i^\dagger P_i F_k \sqrt{\rho} &= U_i^\dagger P_i^\dagger F_j P \sqrt{\rho} \\
&= \frac{U_i^\dagger U_i P F_i^\dagger F_j P \sqrt{\rho}}{\sqrt{d_{ii}}} \\
&= \delta_{ij} \sqrt{d_{ij}} \sqrt{\rho}
\end{aligned}
$$

Thus, one can easily check that this recovery operation satisfies the error correction condition:

$$
\begin{aligned}
(\mathcal{R} \circ \mathcal{E})(\rho) &= \sum_{i,j} U_i^\dagger P_i F_j \rho F_j^\dagger P_i U_i \\
&= \sum_{i,j} \delta_{ij} d_{ii} \rho \propto \rho
\end{aligned}
$$

($\Rightarrow$) Assume $\{E_i\}$ are errors correctable by $\mathcal{R}$, with operation elements $\{R_j\}$. We define a quantum operation $\mathcal{E}_C$ by $\mathcal{E}_C(\rho) = \mathcal{E}(P\rho P)$. $P\rho P \in C$ for any ρ, so we can apply the error correction condition to obtain:

$$
\mathcal{R}(\mathcal{E}_C(\rho)) \propto P\rho P
$$

We can rewrite the left-hand side of this equation as:

$$
\mathcal{R}(\mathcal{E}_C(\rho)) = \sum_{i,j} R_j E_i P \rho P E_i^\dagger R_j^\dagger
$$

Thus

$$
\sum_{i,j} R_j E_i P \rho P E_i^\dagger R_j^\dagger = c P \rho P
$$

for some constant c. The quantum operation with operation elements $\{R_j E_i\}$ is identical to the quantum operation with only one operation element $\sqrt{c}P$. This implies that there exist a complex matrix with entries c_{ki} such that $R_k E_i P = c_{ki} P$. We take the adjoint and obtain $P E_i^\dagger R_k^\dagger R_k E_j P = c_{ki}^* c_{kj} P$. Since $\mathcal{R}$ is a CPTP map, we have $\sum_k R_k^\dagger R_k = I$. Thus, we can sum over k to obtain:

$$P E_i^\dagger E_j P = \alpha_{ij} P$$

Which is the Knill-Laflamme condition, with $\alpha_{ij} = \sum_k c_{ki}^* c_{kj}$.

□

The proof of the Knill-Laflamme condition explicitly provides a method to construct the recovery operation $\mathcal{R}$, given the code subspace and the noise model. However, a quantum system could be affected by a noise process that is not known. It would be desirable to have a code C and an error-correction operation $\mathcal{R}$ that can correct the effect of an entire class of noise processes. That is indeed possible, and it is a consequence of the linearity of quantum mechanics. If we have a code that can correct the effect of a set of errors $\{E_i\}$, then it can also correct the effect of any error that can be expressed as a linear combination of those errors. This is a direct consequence of theorem 1, since if C satisfies the Knill-Laflamme condition for $\{E_i\}$, then it also satisfies it for any linear combination of those errors.

1.3 Stabilizer Codes

The previous discussion has proven the theoretical possibility of quantum error correction. However, the state-vector representation of quantum states carries a practical impossibility. When we have a a large number n of physical qubits, the manipulation of the 2^n-dimensional state vector becomes unfeasible. We need a more efficient way to represent quantum states, and the stabilizer formalism provides us with such a tool, as well as a method to construct QECC.

1.3.1 Stabilizer Formalism

If we take a closer look at the Shor code, and specifically at the syndrome measurement, we can see that in order to detect a bit flip on one of the first three qubits, we measured the eigenvalues of the operators $Z_1 Z_2$ and $Z_2 Z_3$. The same procedure should be performed for the other two blocks of three qubits. In order to detect phase-flip errors, we need to measure the eigenvalues of the operators $X_1X_2X_3X_4X_5X_6$ and $X_4X_5X_6X_7X_8X_9$. In total, we need to measure the eigenvalues of eight operators.The two valid codewords, $|0_L\rangle$ and $|1_L\rangle$, are the eigenstates of all these operators with eigenvalue $+1$, and hence are simulatenously fixed by any of them. Those are called stabilizer operators, and the basic idea of the formalism is to describe the code space as the subspace stabilized by these operators.

In order to give a more precise description of the stabilizer formalism, let's introduce the n-dimensional Pauli group $\mathcal{P}_n$, which is the group generated by the n-fold tensor products of the Pauli matrices and the identity, together with the multiplicative factors ± 1 and $\pm i$. For instance, the 1-dimensional Pauli group is

$$\mathcal{P}_1 = \{\pm I, \pm X, \pm Y, \pm Z, \pm iI, \pm iX, \pm iY, \pm iZ\}$$

It is easily verifiable that $\mathcal{P}_n$ is a group under matrix multiplication. The stabilizer is some Abelian subgroup $\mathcal{S} \subseteq \mathcal{P}_n$ that does not contain $-I$, and the vectors fixed by all elements of $\mathcal{S}$ form the code space $C(\mathcal{S})$, i.e.:

$$C(\mathcal{S}) \equiv \{|\psi\rangle \in \mathcal{H} : g\,|\psi\rangle = |\psi\rangle\,;\ \forall g \in \mathcal{S}\}$$

It is also obvious that the code space $C(\mathcal{S})$ is indeed a vector space. $\mathcal{S}$ is necessarily an Abelian subgroup, since two non-conmuting operators cannot have common eigenstates.

Let's recall that our aim is to define a subspace C of the total Hilbert space, such that it encodes k logical qubits into n physical qubits. Therefore, the dimension of C must be 2^k. We want to see how this condition translates on the definition of the stabilizer $\mathcal{S}$:

Proposition 1. *If $\mathcal{S}$ is an Abelian subgroup of $\mathcal{P}_n$ that does not contain $-I$, then the code space $C(\mathcal{S})$ stabilized by $\mathcal{S}$ has dimension 2^{n-k}, where k is the number of independent generators of $\mathcal{S}$.*

Proof. Let $\mathcal{S}$ be generated by $\langle g_1, g_2, \ldots, g_k\rangle$. Since all the generators are elements of $\mathcal{P}_n$, they are unitary and Hermitian, and thus have eigenvalues ± 1. We can project the total Hilbert space into the $+1$ eigenspace of each generator g_i by applying the projector

$$P_i = \frac{I + g_i}{2}$$

The code space $C(\mathcal{S})$ is the subspace stabilized by all the generators, and thus can be obtained by applying the projectors associated with each generator to the total Hilbert space:

$$P_C = \prod_{i=1}^{k} P_i = \prod_{i=1}^{k} \frac{I + g_i}{2}$$

If we expand the product, we sum over all possible products of the generators g_i, which yields the 2^k elements of $\mathcal{S}$.

$$P_C = \frac{1}{2^k} \sum_{g \in \mathcal{S}} g$$

The dimension of the code space $C(\mathcal{S})$ is given by the trace of the projector P_C:

$$\dim(C(\mathcal{S})) = \mathrm{Tr}(P_C) = \frac{1}{2^k} \sum_{g \in \mathcal{S}} \mathrm{Tr}(g)$$

We have the following cases for the trace of an element $g \in \mathcal{S}$:

- If $g = I^{\otimes n}$, then $\mathrm{Tr}(g) = 2^n$.

- If $g \neq I^{\otimes n}$, then g is a non-trivial element of $\mathcal{P}_n$, and thus has trace zero, since it contains at least one Pauli matrix that has zero trace.

$-I \notin \mathcal{S}$ ensures that there are no elements of $\mathcal{S}$ with trace -2^n. Thus, the only element of $\mathcal{S}$ with non-zero trace is the identity, and we have:

$$\dim(C(\mathcal{S})) = \mathrm{Tr}(P_C) = \frac{1}{2^k} \cdot 2^n = 2^{n-k}$$

$\square$

Therefore, a code that encodes k logical qubits into n physical qubits can be defined by a stabilizer $\mathcal{S}$ with $n-k$ independent generators. This proposition also provides a reason for the non-inclusion of $-I$ in the stabilizer. This is nonetheless obvious from the fact that the only solution to $-I\left|\psi\right\rangle = \left|\psi\right\rangle$ is the trivial solution $\left|\psi\right\rangle = 0$.

1.3.2 Unitary Dynamics of Stabilizer Codes and the Gottessman-Knill Theorem

We have discussed the description of a subspace using stabilizer formalism. Now let's see how we can describe the dynamics of those subspaces under the action of noise and other quantum processes. Let $\left|\psi\right\rangle$ be an element of a code space $C(\mathcal{S})$ stabilized by $\mathcal{S}$. If we apply a unitary operation U,

$$U\left|\psi\right\rangle = Ug\left|\psi\right\rangle = (UgU^\dagger)U\left|\psi\right\rangle$$

and we see that $U\left|\psi\right\rangle$ is stabilized by the conjugated operator $UgU^\dagger$. Since this is common to any element of $C(\mathcal{S})$, we can easily see that $UC(\mathcal{S})$ is stabilized by $USU^\dagger \equiv \{UgU^\dagger : g \in \mathcal{S}\}$. This property does not ensure that any stabilizer of a subspace $UC(\mathcal{S})$ is a valid stabilizer, since the conjugation by U might introduce elements that are not in $\mathcal{P}_n$. We hence introduce the Clifford group $\mathcal{C}_n$ as the normalizer of the Pauli group:

$$\mathcal{C}_n \equiv \{U : U\mathcal{P}_nU^\dagger = \mathcal{P}_n\}$$

The Clifford group is the largest group of unitary operations that preserve the Pauli group under conjugation. If we apply an operation $U \in \mathcal{C}_n$ to a code space C stabilized by $\mathcal{S}$, the resulting subspace UC is stabilized by $USU^\dagger \in \mathcal{P}_n$. Conjugation by a Clifford operation also preserves the commutation relations

$$Ug_1U^\dagger Ug_2U^\dagger = Ug_1g_2U^\dagger = Ug_2g_1U^\dagger = Ug_2U^\dagger Ug_1U^\dagger$$

which ensures that the resulting stabilizer $USU^\dagger$ is also an Abelian subgroup of $\mathcal{P}_n$. If $-I$ was in $U\mathcal{S}U^\dagger$, then there would be some $g \in \mathcal{S}$ such that $UgU^\dagger = -I$, and therefore:

$$\begin{gathered} U^\dagger(UgU^\dagger)U = U^\dagger(-I)U \\ g = -I \end{gathered}$$

which contradicts the definition of stabilizer. Thus, $-I$ is not in $U\mathcal{S}U^\dagger$, and the resulting subspace UC is a valid code space stabilized by $USU^\dagger$.

The Clifford group contains the Pauli group, but is larger. $\mathcal{C}_n$ is generated by the Hadamard gate, the phase gate, and the CNOT gate. It is important to note that this small set is enough to create highly entangled states.

This formalism provides us with a method to efficiently represent the dynamics of quantum circuits under the action of Clifford operations. The idea is that, instead of representing the state of the system as a vector in a 2^n-dimensional Hilbert space, we can represent it by its stabilizer, which is generated by n independent generators. Each generator is an element of $\mathcal{P}_n$, and thus can be represented by a string of n characters, each of which can be I, X, Y, or Z. Therefore, we can represent the state of the system by a list of n strings of length n. This discussion culminates in the following theorem:

Theorem 2 (Gottesman-Knill)**.** *Any quantum circuit that consists solely of Clifford gates, preparation of qubits in the computational basis, and measurements of Pauli operators can be efficiently simulated on a classical computer.*

The Clifford group is not universal for quantum computation, and thus the Gottesman-Knill theorem does not contradict the fact that quantum computers can outperform classical computers. However, it is important to note that the Clifford group is a very large group of operations, and it is enough to create highly entangled states, and many quantum error correction circuits, as well as quantum protocols such as teleportation or superdense coding can be implemented using only Clifford gates. In order to achieve universality, we need to add at least one non-Clifford gate, such as the T gate.

1.3.3 Error Detection and Correction in Stabilizer Codes

Now that we have established the stabilizer formalism, we can explicitly describe how a stabilizer code detects and corrects errors in a remarkably elegant way. [5]

Let $C(\mathcal{S})$ be an $[n, k]$ stabilizer code defined by the stabilizer group $\mathcal{S} = \langle g_1, g_2, \ldots, g_{n-k} \rangle$. A valid encoded state $|\psi_L\rangle \in C(\mathcal{S})$ satisfies $g_i |\psi_L\rangle = |\psi_L\rangle$ for all $i \in \{1, \ldots, n-k\}$.

Suppose an arbitrary error occurs. As discussed in Section 1.1, we can discretize errors by expanding them in the basis of the Pauli group $\mathcal{P}_n$. Thus, we only need to consider what happens when a Pauli error $E \in \mathcal{P}_n$ corrupts our state, transforming it into $E|\psi_L\rangle$. To detect the error without measuring the logical state (which would destroy the quantum information), we measure the $n-k$ stabilizer generators g_i.

Since both E and g_i belong to the Pauli group $\mathcal{P}_n$, they must either commute ($Eg_i = g_iE$) or anti-commute ($Eg_i = -g_iE$). This fundamental property of the Pauli group dictates the outcome of our syndrome measurements:

- If E commutes with g_i: $g_i(E|\psi_L\rangle) = Eg_i|\psi_L\rangle = E|\psi_L\rangle$. The corrupted state is an eigenstate of g_i with eigenvalue $+1$.
- If E anti-commutes with g_i: $g_i(E|\psi_L\rangle) = -Eg_i|\psi_L\rangle = -E|\psi_L\rangle$. The corrupted state is an eigenstate of g_i with eigenvalue -1.

The outcomes of these $n-k$ measurements form the *error syndrome*. Typically, we map the eigenvalues ± 1 to binary values $s_i \in \{0, 1\}$, where $s_i = 0$ corresponds to

+1 (commutes) and $s_i = 1$ corresponds to -1 (anti-commutes). The syndrome vector $\vec{s} = (s_1, s_2, \ldots, s_{n-k})$ provides a unique fingerprint for the error E, provided that the code is capable of correcting it.

Once the syndrome $\vec{s}$ is extracted, classical processing (decoding) is used to determine the most likely error E that occurred. The correction step simply consists of applying a recovery operator $R \in \mathcal{P}_n$ that shares the same syndrome as E. If the decoding is successful, RE will commute with all stabilizers, and ideally $RE \in \mathcal{S}$. Since elements of $\mathcal{S}$ act as the identity on the code space, the combined effect of the error and the recovery operation is trivial, successfully restoring the original state: $RE\,|\psi_L\rangle = |\psi_L\rangle$.

However, not all errors can be successfully corrected. If an error E commutes with all stabilizer generators but is not an element of $\mathcal{S}$, it will yield a trivial syndrome (all +1) while transforming the state into a different, orthogonal codeword. The set of all operators in $\mathcal{P}_n$ that commute with all elements of $\mathcal{S}$ is the centralizer of $\mathcal{S}$, which for the Pauli group is equivalent to the normalizer $N(\mathcal{S})$.

Therefore, undetectable and uncorrectable logical errors are exactly those in $N(\mathcal{S}) \setminus \mathcal{S}$. The minimum weight (defined as the number of non-identity tensor factors in the Pauli string) of an operator in $N(\mathcal{S}) \setminus \mathcal{S}$ defines the distance d of the code. Such a code is usually denoted by the parameters $[n, k, d]$, meaning it encodes k logical qubits into n physical qubits and can correct any arbitrary error affecting up to $\lfloor (d-1)/2 \rfloor$ physical qubits.

1.4 Fault-Tolerant Quantum Computation

We have discussed how encoding quantum information into a subspace can protect the data from environmental noise. However, the very process of encoding, syndrome measurement, as well as the application of logical quantum gates can also be affected by noise. The fundamental idea of fault-tolerant quantum computation is to design these processes in such a way that they do not propagate errors uncontrollably, and that the error correction procedure can still successfully correct the errors that occur during these operations.

Multi-qubit gates carry the risk of propagating errors from one physical qubit to another. The reiterative application of such operations can produce a single physical hardware failure to cascade into a large number of errors on the physical qubits encoding the logical state. If the weight of the resulting error exceeds the distance d of the code, the error correction procedure will fail. Therefore, the design of fault-tolerant quantum circuits must ensure that errors do not propagate in a way that exceeds the error-correcting capabilities of the code.

We can prevent catastrophical error propagation by imposing the condition of transversality on the implementation of logical gates. A transversal gate is one that can be implemented by applying independent physical operations exclusively between corresponding physical qubits in different code blocks. This architectural method ensures that a single physical error propagates to at most one physical qubit in each code block, thus maintaining the error weight within the correctable treshold of the code. This may seem like a very simple condition, but it has a profound limitation [3]:

Theorem 3 (Eastin-Knill)**.** *No local quantum error-correcting code can implement a universal set of logical gates transversally.*

This theorem implies that the implementation of transversal gates alone does not suffice to achieve universal fault-tolerant quantum computation.

2 Topological Quantum Error Correction

We have introduced the algebraic framework for diagonsing and correcting quantum errors. However, the practical implementation of QECC must face another critical challenge: the strict near-neighbor connectivity of physical qubits in real scalable hardware platforms, such as superconducting circuits [1] [6]. This constraint necessitates the development of quantum error-correcting codes that can be implemented with local interactions, i.e., families of stabilizer codes where the number of qubits in the support of any of the stabilizer generators must be bounded. This will be the depart point of topological quantum codes, a concept initiated by Kitaev's toric code [7].

2.1 Homology

We are interested in a specific family of stabilizer codes, namely surface codes. Those can be regarded as encoding quantum information in the homological degrees of freedom of a two-dimensional lattice. In order to understand the construction of surface codes, we need to introduce some concepts from algebraic topology [2].

Let's consider a connected, closed, orientable surface of genus g, and define a lattice on it, with V vertices. E edges and F faces. We will call 0-cells the vertices of the lattice, 1-cells the edges, and 2-cells the faces.

We want to obtain the homology groups of this surface over $\mathbb{Z}_2$. For this purpose, we construct an abelian group for each dimension $i \in \{0, 1, 2\}$, generated by the sets of i-cells, where the field of coefficients is $\mathbb{Z}_2$. In this sense, if we consider a set of 1-cells $E = \{e_1, e_2, \dots, e_m\}$, we can represent it as a sum

$$c = \sum_i c_i e_i \qquad c_i = \begin{cases} 0 & e_i \notin E \\ 1 & e_i \in E \end{cases}.$$

Such sums are called 1-chain. We can consider addition of 1-chains (taking into account that the coefficients are in $\mathbb{Z}_2$ and hence $e_i + e_i = 0$), obtaining an abelian group structure on the set of 1-chains C_1. The elements of C_1 can be considered as binary vectors of length E, and topologically $C_1 \simeq \mathbb{Z}_2^E$. Equivalently, we define the group of 0-chains $C_0 \simeq \mathbb{Z}_2^V$ as the abelian group generated by the sets of 0-cells, and the group of 2-chains $C_2 \simeq \mathbb{Z}_2^F$ using the exact same procedure for 2-cells.

The entire homology theory rests on the definition of the boundary operators $\partial_i : C_i \to C_{i-1}$, which are a family of group homomorphisms that take an i-chain and return its geometric boundary as an $(i-1)$-chain. Given our construction, we are interested in two boundary operators:

- $\partial_1 : C_1 \to C_0$ takes a 1-chain (a set of edges) and returns the set of vertices that are the endpoints of an odd number of edges in the 1-chain. For instance, if we have a 1-chain $c = e_1 + e_2 + e_3$, where e_1 connects vertices v_1 and v_2, e_2 connects vertices v_2 and v_3, and e_3 connects vertices v_3 and v_4, then $\partial_1(c) = v_1 + v_4$, since v_1 and v_4 are the endpoints of an odd number of edges in c, while v_2 and v_3 are the endpoints of an even number of edges in c.
- $\partial_2 : C_2 \to C_1$ takes a 2-chain (a set of faces) and returns the set of edges that are the endpoints of an odd number of faces in the 2-chain.

The boundary operators satisfy the property that they are nilpotent of degree 2, meaning that $im(\partial_{i+1}) \subseteq ker(\partial_i)$, or equivalently $\partial_i \circ \partial_{i+1} = 0$. This property is a direct consequence of the geometric interpretation of the boundary operators, since the boundary of a boundary is always empty.

From these boundary operators, we construct two subgroups of C_i. The group of i-cycles Z_i is defined as the kernel of ∂_i, that is, the set of i-chains that have an empty boundary. The group of i-boundaries B_i is defined as the image of ∂_{i+1}, that is, the set of i-chains that are the boundary of some $(i+1)$-chain. Since $\partial_i \circ \partial_{i+1} = 0$, we have that $B_i \subseteq Z_i$.

We can define the i-th homology group as the algebraic quotient group of i-cycles by i-boundaries:

$$H_i = \frac{Z_i}{B_i}$$

where the quotient group is defined as the set of equivalence classes of i-cycles, where two i-cycles are equivalent if they differ by a i-boundary, that is, the elements of H_i are of the form $[z] = \{z + b : b \in B_i\}$, where z is any i-cycle. The homology groups are topological invariants, meaning that they do not depend on the specific lattice we have defined on the surface, but only on the topology of the surface itself. For a connected, closed, orientable surface of genus g, we have that $H_0 \simeq \mathbb{Z}_2$, $H_1 \simeq \mathbb{Z}_2^{2g}$ and $H_2 \simeq \mathbb{Z}_2$.

2.2 Surface Codes

Consider a lattice embedded in a closed surface, and attach a qubit to each edge. It is natural to see elements of the computational basis as 1-chains $c \in C_1$

$$|c\rangle = \bigotimes_i |c_i\rangle$$

It is useful to define groups homeomorphism from C_1 to the group of Pauli operators $\mathcal{P}_n$, where n is the number of edges in the lattice and hence the number of physical qubits.

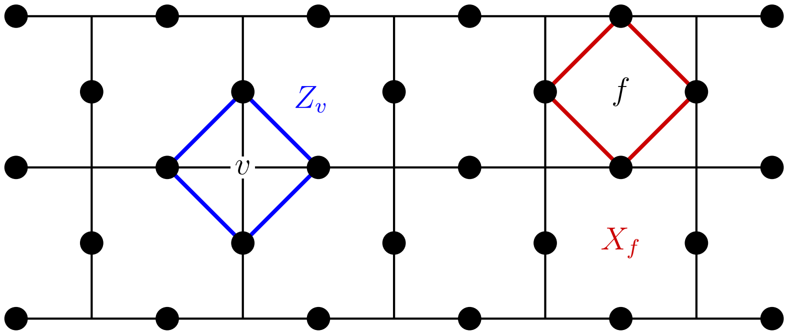

Figure 2: Lattice showing the arrangement of physical qubits and the support of the stabilizer generators.

We can define two homomorphisms, $X : C_1 \to \mathcal{P}_n$ and $Z : C_1 \to \mathcal{P}_n$, such that

$$X_c = \bigotimes_i X_i^{c_i} \quad \text{and} \quad Z_c = \bigotimes_i Z_i^{c_i}.$$

Given these homomorphisms, we can define the face and vertex operators as follows:

$$X_f := \prod_{e \in \partial_2 f} X_e \quad \text{and} \quad Z_v := \prod_{e | v \in \partial_1 e} Z_e$$

Those operators are ilustrated in figure 2, giving a clear intuition of their geometric meaning. The face and vertex operators commute with each other, and hence we can consider a stabilizer group $\mathcal{S}$ generated by them. In order to see the code subspace stabilized by $\mathcal{S}$, we can apply the projectors associated with each generator to any state of the total Hilbert space $|c\rangle$, $c \in C_1$:

$$\prod_f \frac{I + X_f}{2} \prod_v \frac{I + Z_v}{2} |c\rangle$$

Let's look at how each of the projectors acts on our arbitrary 1-chain state $|c\rangle$. The vertex operator Z_v effectively checks the parity of the edges of the chain c connected to the vertex v:

$$Z_v |c\rangle = (-1)^k |c\rangle \quad \text{where } k = |\{e \mid v \in \partial_1 e,\ e \in c\}|$$

From the definition of the boundary operator ∂_1, we can see that $Z_v |c\rangle = |c\rangle$ if and only if v is the endpoint of an even number of edges in c and hence $v \notin \partial_1 c$. If $v \in \partial_1 c$, then $Z_v |c\rangle = -|c\rangle$ and the projector $(I + Z_v)|c\rangle = 0$. This means that the only states that survive the application of all vertex projectors are those associated with 1-cycles, $|z\rangle$, with $z \in Z_1$.

Now we apply the face projectors to our surviving states $|z\rangle$:

$$\prod_f \frac{1 + X_f}{2} |z\rangle$$

We can expand the product of all face terms and obtain a sum over all posible subsets of faces $\{f_i\}$. We also use that applying face operators on adjacent faces cancels out the operators on their shared edges:

$$\prod_f (1 + X_f) = \sum_{\{f_i\}} \prod_i X_{\partial_2 f_i} = \sum_{\{f_i\}} X_{\partial_2(\sum_i f_i)}$$

We notice that $\sum_i f_i$ is a 2-chain, and thus $\partial_2(\sum_i f_i)$ is a 1-boundary. Therefore this is proportional to the sum over all possible 1-boundaries:

$$\prod_f (1 + X_f) \propto \sum_{b \in B_1} X_b$$

Applying this to our cycle state $|z\rangle$ yields:

$$\prod_f \frac{1 + X_f}{2} |z\rangle \propto \sum_{b \in B_1} X_b |z\rangle = \sum_{b \in B_1} |z + b\rangle$$

This means that the states associated with two different cycles differing by a boundary are projected to the same state, and hence the code subspace stabilized by $\mathcal{S}$ is the vector space generated by the states $|[z]\rangle = \sum_{b \in B_1} |z + b\rangle$, where $[z]$ is the equivalence class of z in H_1. Since $H_1 \simeq \mathbb{Z}_2^{2g}$, we have that the code subspace has dimension 2^{2g}, and hence encodes $k = 2g$ logical qubits into n physical qubits.